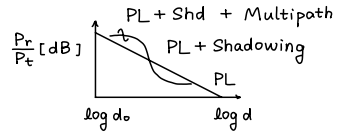
$c = \lambda f$ $\quad 1\,m/s = 3.6\,km/h$ $\quad 1000\,mW = 1W = 0\,dB = 30\,dBm$ $\quad [dB] = 10\lg[W]$

$G: 10^9$ $\quad M$ (Mega) $: 10^6$ $\quad n: 10^{-9}$ $\quad \mu$ (micro) $: 10^{-6}$ $\quad m$ (milli) $: 10^{-3}$ $\quad Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$

2G: GSM. Analog -> Digital Equalization Drawback: High complexity High SINR

3G: CDMA IS-95 Drawback: low capacity, low spectrum efficiency

4G: LTE. 2 tech: OFDMA. MIMO Drawback: time delay




Fading: Time-variation of channel due to $\begin{cases} \text{small scale by multipath} \\ \text{larger scale} \end{cases}$

Path Loss $PL[dB] = P_t[dBm] - P_r[dBm]$ $\quad \begin{cases} \text{Path Loss by long distance} \\ \text{Shadowing by obstacle} \end{cases}$

Free-Space $P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$ Far-field dist $d > \frac{2D^2}{\lambda}$

Simplified Path $PL \propto \left(\frac{d}{d_0}\right)^n$. $PL(d)[dBm] = K[dBm] - 10n\lg\frac{d}{d_0} = PL(d_0)[dBm] + 10n\lg\frac{d}{d_0}$

$P_r[dBm] = P_t[dBm] + K[dB] - 10n\lg\frac{d}{d_0}$ $\quad K[dB] = 20\log\frac{\lambda}{4\pi d_0}$

Shadowing Fading $PL(d)[dBm] = PL(d_0)[dBm] + 10n\lg\frac{d}{d_0} + \psi_{dB}$ $\quad \psi_{dB} \sim N(0, \sigma^2)$

MMSE $\min_{K,n} \Sigma(PL_{测} - K + 10n\lg d)^2$ Outage Probability $P_{out} = P(P_r(d) < P_{min}) = 1 - Q\left(\frac{P_{min}}{\sigma}\right)$

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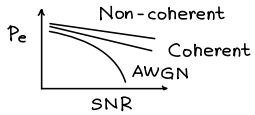



Rayleigh Fading: If no LOS, Amplitude of signal after multipath propagation

$|h| = \sqrt{Re h^2 + Im h^2}$ PDF $\frac{x}{\sigma^2} e^{-x^2/2\sigma^2}$ $|h|^2 \sim Exp(\sigma^2)$ Racian Fading: If there exists a strong LOS

$y[m] = \sum_l h_l[m]\, x[m-l]$ Flat $y[m] = h[m]\, x[m]$ Slow $y[m] = \sum_l h_l\, x[m-l]$

Delay Spread $T_d = \max|\tau_i(t) - \tau_j(t)|$ Coherence Bandwidth $W_c = \frac{1}{2T_d}$ $\quad W$ Basebandwidth

Doppler Shift $D = \frac{f_c v}{c}$ Time-scale for change of path phase $\frac{1}{4D}$ $\quad T_s$ Symbol Duration

Doppler Spread $D_s = \max_{i,j} f_c|\tau_i'(t) - \tau_j'(t)|$ Doppler Shift of ith path $f_c\tau_i'(t)$, $\tau_i'(t) = \frac{v}{c}\cos\theta$

Coherence Time $T_c = \frac{1}{D_s}$ Time-scale for change of path amplitude $d/v$ Underspread $T_d \ll T_c$

Time-scale for a path to move over a tap $c/(vW)$ $\quad \begin{cases} \text{Flat } W_c \gg W,\ T_d \ll T_s \\ \text{Selective } W_c < W,\ T_d > T_s \end{cases}$

Constellation Energy $\mathcal{E} = \frac{1}{M}\sum_{n=1}^{M} E|s(n)|^2$ QPSK $\frac{1}{\sqrt{2}}$ 16-QAM $\frac{3}{\sqrt{10}}$

AWGN $y = x + w$. BPSK $x = \pm a$. $SNR = a^2/N_0$. $P_e = Q(\sqrt{2SNR})$ $\propto e^{-SNR}$ $\quad \begin{cases} \text{Slow } D_s \ll W,\ T_c \gg T_s \\ \text{Fast } D_s > W,\ T_c < T_s \end{cases}$

Rayleigh Channel $y = hx + w$, $h \sim N(0,1)$

Know h, Coherence Direction $r = Re\frac{h^*}{|h|}y = |h|x + z$, $z \sim N(0, \frac{N_0}{2})$ $\quad P_e = Q(\sqrt{2|h|^2 SNR})$

Diversity: ensure the symbols pass through multipath signal paths each of which fades indply

TDMA Repetition Code $y_l = h_l x + w_l$ Match Filter / MRC $\tilde{y} = \frac{1}{\|h\|}h^H y = \|h\|x + \tilde{w}$ $\quad \|h\|^2 \sim \chi^2(2N)$

Rotation Code, QPSK Normalized codeword distance $d = 2\begin{vmatrix}\cos\theta \\ \sin\theta\end{vmatrix}$ Product distance $|d_1 d_2|$

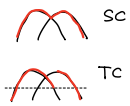
$y = u + w = h \odot x + w$ $\quad Pr(x_A \to x_B) = Q\left(\frac{\|u_A - u_B\|/2}{\sqrt{N_0/2}}\right) \le \frac{16}{|d_1 d_2|^2} SNR^{-2}$

Receiver Combining Technique

Measured SNR $\gamma \sim Exp(\frac{1}{\bar{\gamma}})$ in Rayleigh Fading PDF $\frac{1}{\bar{\gamma}} e^{-\gamma/\bar{\gamma}}$ $\quad$ CDF $1 - e^{-\gamma/\bar{\gamma}}$ Average SNR $\bar{\gamma} = \frac{E_s}{N_0}$

Outage Probability $P_{out} = Pr(\gamma < \gamma_0) = 1 - e^{-\frac{\gamma_0}{\bar{\gamma}}}$

SC (Selection Combining) $\gamma_\Sigma = \max \gamma_m$ CDF $Pr(\gamma_\Sigma < \gamma) = \prod_{m=1}^{M} Pr(\gamma_m < \gamma) = (1 - e^{-\gamma/\bar{\gamma}})^M$




$P_{out} = P_{\gamma_\Sigma}(\gamma_0) = (1 - e^{-\gamma_0/\bar{\gamma}})^M$ $\quad \bar{\gamma}_\Sigma = \int \gamma\, p_{\gamma_\Sigma}(\gamma)\, d\gamma = \bar{\gamma}\cdot\sum_{m=1}^{M}\frac{1}{m}$. $\quad A_g = \sum_{m=1}^{M}\frac{1}{m}$

TC (Threshold Combining): The 1-st above threshold is used

MRC (Maximum Ratio Combining) $y_m = h_m x + w_m$. $\tilde{y} = \frac{1}{\|h\|}h^H y$. If $h_m = r_m e^{j\theta_m}$,

$\gamma_\Sigma = \sum_{m=1}^{M}\gamma_m \sim \chi^2(2M)$. $A_g = M$ $\quad$ weight $\alpha_m = a_m e^{-j\theta_m}$. $a_m = r_m/\sqrt{N_0}$

EGC (Equal Gain Combining) $\alpha_m = e^{-j\theta_m}$ $\quad A_g = [1 + (M-1)\frac{\pi}{4}]$

Trasmitter Diversity Known CSI (Channel State Information): MRC at Tx. $x_m = a_m e^{-j\theta_m} x$

Unknown CSI : Alamouti Scheme / OTSBC At time 1, transmit $x_1, x_2$ Each transmission $E_s/2$

At time 2, transmit $-x_2^*, x_1^*$ ( since 2 transmitters)

$$\begin{bmatrix} y[1] \\ y^*[2] \end{bmatrix} = \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} w[1] \\ w^*[2] \end{bmatrix} \qquad z = H^H y = H^H H x + H^H w = (|h_1^2| + |h_2^2|) x + \tilde{w}$$

SNR $\gamma_1 = \gamma_2 = (|h_1^2| + |h_2^2|) \frac{E_s}{2N_0}$ $H_{M\times L}$ elements has P choices Transmitter M, Symbol P, Time Slot L, Rate R

Diversity = 2, $A_g = 1$, 2 DoF, 2 × rate of repeating coding. 3 dB loss ( half ) SNR to trsmitter bf

Space - Time code design set of $\{X_i\}$ Coding Gain <- determinants of $(X_i - X_j)(X_i - X_j)^H$

Full diversity is achieved if all pairwise differences has full rank time diversity $X_i = \mathrm{diag}(x_i)$

MIMO 2×2 $y = Hx + w = [h_1 \ h_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + w$. ML high complexity.

Sub-optimal scheme : zero-forcing $H^{-1}y$. $y_1' = (\phi_2^H h_1) x_1 + z_1$, $h_2 \perp \phi_2$

Channel inverse elimates the interference from Tx 2 by proj $y$ onto $\phi_2$. price : reduce diversity gain

Selective Fading, ISI, $y[m] = \sum_{l=0}^{L-1} h_l x[m-l] + w[m]$

2G, GSM : Equalization compensate signal distortion

$\Psi$ x[0] x[1] x[2]
$\Psi$ x[0] x[1]
$\Psi$ x[0]

MISO $y = h^H X_{L\times(N+L-1)} + w$ L : trsmt / delay time slot N : codelengh

MLSD ( ML Sequence Detection) : max $p(y|x)$ Viterbi Algo : the memory of selective fading be captured in finite state machine, solve shortest path problem

3G, CDMA Raker Receiver

DSSS ( Direct Sequence Spread Spectrum ) : $s(t) *$ pseudo-random spreading code seq

transmit in multipath then MRC receive

IDFT → CP → Channel → × CP → DFT

OFDM for selective fading, ISI, Parallel Channel

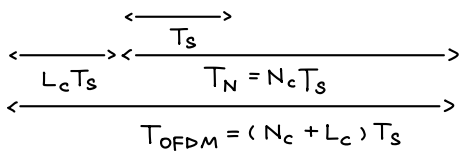
Subcarrier $N_c$ Symbol Period / Sample interval $T_s$ Cyclix Perfix $L_c$ Guard interval $L_cT_s$

Subcarrier Spacing $\Delta = \frac{W}{N_c} = \frac{1}{T_N}$ Total Bandwidth $W = \frac{1}{T_s}$ Message interval $T_N = N_cT_s$

OFDM Period $T_{OFDM} = (N_c + L_c)T_s$ Overhead $L_c / L_c + N_c$ No ISI $T_d \ll L_cT_s$

Rate $R_{min} = \log_2 M \cdot r_{code} \cdot N_c / T_{OFDM}$ No Doppler $T_c \gg T_{OFDM}$

OFDMA is built on OFDM allow multi-user share the spectrum

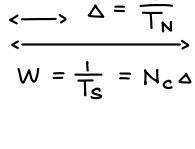



Cellular Network : Hexagonal, Freq Reuse

Channels S, Cells in a cluster N, Reuse factor $\frac{1}{N}$

Allocated channels in a cell $k = S/N$, Cluster number M

N ↓, C ↑, Interference ↑ Capacity $C = MS$ Cell Splitting, Sectoring




AWGN $C = \log_2(1 + SNR)$ bit / s / Hz Continous time $C = W\log(1 + \frac{P}{N_0W})$ bit / s

SIMO, MRC / MISO FullCSI $C = \log(1 + \|h\|^2 SNR)$

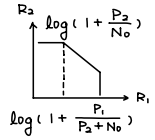
Selective Fading -> OFDM Waterfilling $C = \sum_{n=1}^{N_c} \log(1 + \frac{|h_n|^2 P_n^*}{N_0})$, $P_n^* = (\frac{1}{\lambda} - \frac{N_0}{|h_n|^2})^+$

Slow Fading with receiver CSI only, $y[m] = hx[m] + w[m]$ $P_{out}(R) = P(\log(1 + |h|^2 SNR) < R)$

receiver diversity $P_{out}(R) = P(\log(1 + \|h\|^2 SNR) < R)$ $= 1 - e^{-\frac{2^R - 1}{SNR}}$

trsmt diversity ( unknown h ) $P_{out}(R) = P(\log(1 + \frac{1}{N_t}\|h\|^2 SNR) < R)$

time diversity $P_{out}(R) = P(\frac{1}{L}\sum_{l=1}^{L} \log(1 + \|h_l\|^2 SNR) < R)$

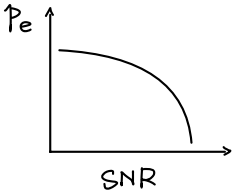



Fast Fading, $y[m] = h[m]x[m] + w[m]$ receiver CSI $C = E[\log(1 + |h|^2 SNR)]$

SIC ( Successive Interference Cancellation ) FullCSI $C = E[\log(1 + |h|^2 \frac{P^*(h)}{N_0})]$

after one user is encoded, its signal is stripped away

Multiuser $C_{sum} = \max_{(R_1, R_2) \in C} R_1 + R_2$ $R_i < \log(1 + \frac{P_i}{N_0})$, $i = 1, 2$ $R_1 + R_2 < \log(1 + \frac{P_1 + P_2}{N_0})$


Pe
SNR